he was right (though he often was not), and no mountain of evidence could convince him to the contrary.

Landau, like Zwicky, had great self-confidence and little fear of appearing a fool. For example, he did not hesitate to publish his 1931 idea that stars are energized by superdense stellar cores in which the laws of quantum mechanics fail. In mastery of theoretical physics, Landau totally outclassed Zwicky; he was among the top ten theorists of the twentieth century. Yet his speculations were wrong and Zwicky's were right. The Sun is *not* energized by neutron cores; supernovae *are* energized by neutron stars. Was Landau, by contrast with Zwicky, simply unlucky? Perhaps partly. But there is another factor: Zwicky was immersed in the atmosphere of Mount Wilson, then the world's greatest center for astronomical observations. And he collaborated with one of the world's greatest observational astronomers, Walter Baade, who was a master of the observational data. And at Caltech he could and 'did talk almost daily with the world's greatest cosmic-ray observers. By contrast, Landau had almost no direct contact with observational astronomy, and his articles show it. Without such contact, he could not develop an acute sense for what things are like out there, far beyond the Earth. Landau's greatest triumph was his masterful use of the laws of quantum mechanics to explain the phenomenon of superfluidity, and in this research, he interacted extensively with the experimenter, Pyotr Kapitsa, who was probing superfluidity's details.

For Einstein, by contrast with Zwicky and Landau, close contact between observation and theory was of little importance; he discovered his general relativistic laws of gravity with almost no observational input. But that was a rare exception. A rich interplay between observation and theory is essential to progress in most branches of physics and astronomy.

And what of Oppenheimer, a man whose mastery of physics was comparable to Landau's? His article, with Volkoff, on the structure of neutron stars is one of the great astrophysics articles of all time. But, as great and beautiful as it is, it "merely" filled in the details of the neutron-star concept. The concept was, indeed, Zwicky's baby—as were supernovae and the powering of supernovae by the implosion of a stellar core to form a neutron star. Why was Oppenheimer, with so much going for him, far less innovative than Zwicky? Primarily, I think, because he declined—perhaps even feared—to speculate. Isidore I. Rabi, a close friend and admirer of Oppenheimer, has described this in a much deeper way: